

End-to-End E-commerce Data Analysis

Google BigQuery | TheLook E-commerce Dataset

1. Project Overview

This project presents an end-to-end e-commerce data analysis conducted using SQL in Google BigQuery on the public TheLook E-commerce dataset. The objective is to identify customer journey friction points and evaluate the website's business performance. The analysis focuses on the **events** table and covers four main areas: conversion funnel analysis, marketing channel performance, user conversion time analysis, and revenue analysis. The findings are then visualized through an interactive dashboard highlighting key KPIs and actionable business insights to support decision-making.

1.1. Technical Environment

- **SSMS (SQL Server Management Studio)** for database management and data exploration
- **Language:** Standard SQL
- **Public Dataset:** bigquery-public-data.thelook_ecommerce
- **Power BI:** Dashboard creation, KPI tracking, and DAX measures

2. Data Processing and Analysis Steps

2.1. Initial Setup and Data Exploration

The first step consists of connecting to the public BigQuery project, loading the TheLook dataset, and exploring the database structure. Initial exploration includes reviewing table schemas, understanding data relationships, and counting records before performing any analytical work.

event_id	user_id	event_type	event_date	product_id	amount	traffic_source
8490	5024	page_view	2025-12-30 04:58:24.5170060	205	NULL	social
2296	1713	page_view	2025-12-30 05:10:26.2763770	201	NULL	organic
3896	2558	page_view	2025-12-30 05:11:09.0018870	404	NULL	organic
2297	1713	add_to_cart	2025-12-30 05:13:26.2763770	201	NULL	organic
2298	1713	checkout_start	2025-12-30 05:16:26.2763770	201	NULL	organic
8438	4993	page_view	2025-12-30 05:20:55.4169290	205	NULL	organic
2299	1713	payment_info	2025-12-30 05:23:26.2763770	201	NULL	organic
2300	1713	purchase	2025-12-30 05:25:26.2763770	201	48.3	organic
8649	5102	page_view	2025-12-30 05:25:27.6001190	102	NULL	organic
3897	2558	add_to_cart	2025-12-30 05:29:09.0018870	404	NULL	organic
2101	1591	page_view	2025-12-30 05:42:46.6687860	305	NULL	organic
3368	2276	page_view	2025-12-30 05:48:25.5165210	201	NULL	organic

2.2. Sales Funnel Analysis

Funnel Stage Definition

The sales funnel is divided into five key stages corresponding to events recorded in the **events** table:

- **Stage 1: Home** — Homepage visit
- **Stage 2: Cart** — Product added to cart

- **Stage 3: Checkout** — Checkout process initiated
- **Stage 4: Payment** — Payment information entered
- **Stage 5: Purchase** — Order successfully completed

For each stage, the number of unique users triggering the event is calculated.

event_type	nb_events
page_view	5000
add_to_cart	1553
checkout_start	1103
payment_info	899
purchase	826

Conversion Rate Calculation

Conversion rates between consecutive stages are calculated as the percentage of users progressing from stage N to stage N+1.

view_to_cart_rate	cart_to_checkout_rate	checkout_to_payment_rate	payment_to_purchase_rate	overall_conversion_rate
31.060000000000	71.020000000000	81.500000000000	91.880000000000	16.520000000000

2.3. Marketing Channel Analysis (Traffic Sources)

The objective is to identify which acquisition channels (Organic Search, Email, Display Advertising, Social Media, Search Ads, etc.) generate the highest traffic volume and which channels achieve the strongest conversion performance through to purchase.

traffic_source	views	carts	view_to_cart_rate	purchases	cart_to_purchase_rate	overall_conversion_rate
organic	2038	669	32.830000000000	343	51.270000000000	16.830000000000
paid_ads	968	358	36.980000000000	204	56.980000000000	21.070000000000
email	522	326	62.450000000000	177	54.290000000000	33.910000000000
social	1472	200	13.590000000000	102	51.000000000000	6.930000000000

2.4. Time-to-Conversion Analysis

This analysis measures the average number of days between a user's first website visit and their first purchase. Understanding this delay helps evaluate customer buying behavior and the length of the purchasing decision cycle.

converted_users	avg_view_to_cart_minutes	avg_cart_to_purchase_minutes	avg_total_journey_minutes
826	11,16	13,47	24,63

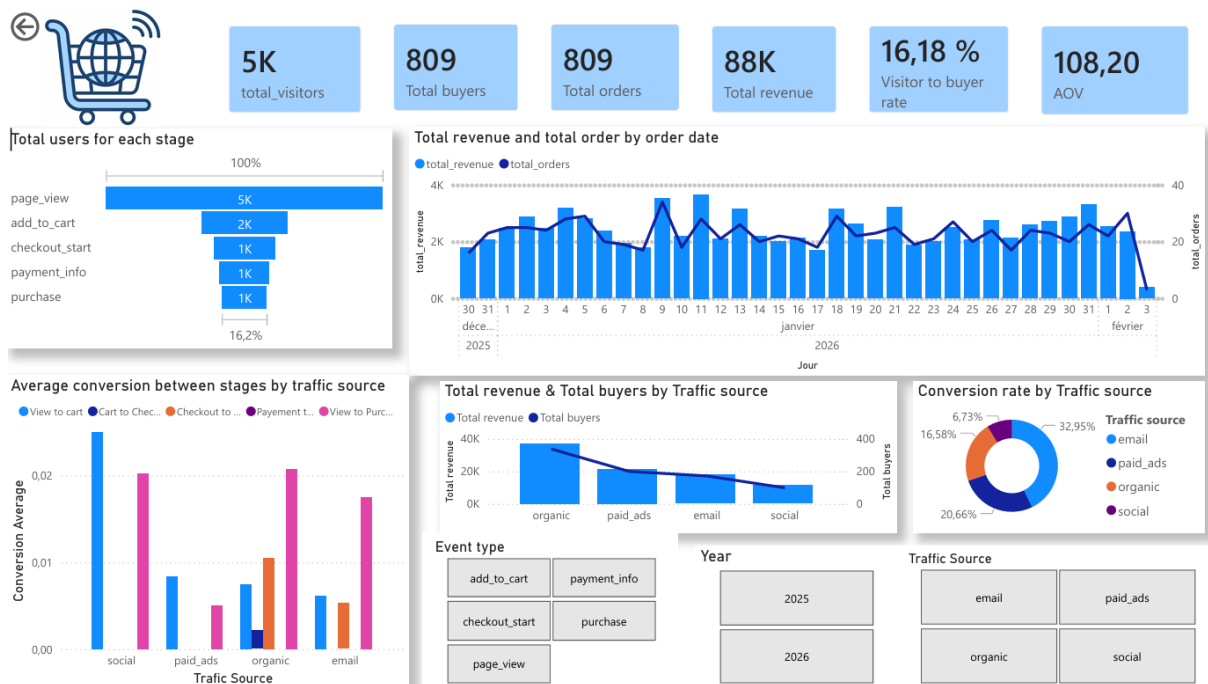
2.5. Revenue Analysis

This section focuses on key financial metrics across the funnel, including:

- Total revenue by traffic source
- Average Order Value (AOV)
- Revenue contribution of each acquisition channel

total_visitors	total_buyers	total_orders	total_revenue	avg_order_value	revenue_per_buyer	revenue_per_visitor
5000	826	826	87975.11	106.510000	106.510000	17.600000

3. Key Insights and Findings



3.1. Conversion Funnel — Friction Points

The funnel analysis reveals significant user drop-off at every stage of the customer journey. While many visitors browse products, only a small percentage complete a purchase.

- The transition from **Home** → **Product Exploration** serves as the first qualification stage, filtering visitors with genuine interest.
- The transition from **Product Exploration** → **Cart** reflects purchase intent and often represents one of the largest drop-off points.
- The transition from **Cart** → **Purchase** is critical, as cart abandonment remains one of the most costly challenges in e-commerce.

Insight: Optimization efforts should primarily focus on reducing cart abandonment through retargeting campaigns, email reminders, and a simplified checkout experience.

3.2. Marketing Channels — Comparative Performance

The marketing channel analysis highlights substantial differences in both traffic volume and traffic quality:

- Organic channels (Organic Search and Email) generally attract users with stronger purchase intent, resulting in higher conversion rates.
- Display advertising often generates large traffic volumes but typically delivers lower conversion performance.
- Email marketing frequently stands out with both higher conversion rates and higher average order values, making it a high-ROI acquisition channel.

Insight: Marketing budgets should be strategically reallocated toward channels with stronger conversion performance, even when traffic volume is lower.

3.3. Time-to-Conversion Analysis

The analysis shows that most conversions occur within the first few days after a user's initial visit. Beyond a certain threshold (typically 7–14 days), the likelihood of conversion decreases significantly.

- A short conversion window (0–2 days) often indicates immediate purchase intent driven by specific searches or promotional offers.
- Longer conversion periods suggest a consideration phase where nurturing emails and remarketing campaigns can influence purchasing decisions.

Insight: Retargeting campaigns and follow-up emails should ideally be triggered within 48 to 72 hours after the user's first visit to maximize effectiveness.

3.4. Revenue Analysis

The revenue analysis identifies the most profitable acquisition channels based on generated revenue and Average Order Value (AOV):

- Some lower-volume channels generate significantly higher AOV, indicating a more qualified and higher-spending customer base.
- Total revenue should not be evaluated in isolation. A channel generating less revenue may still provide superior ROI due to a lower Customer Acquisition Cost (CAC).
- Monthly revenue trends reveal seasonal peaks linked to events such as sales promotions and holiday periods, enabling better forecasting and planning.

4. Business Recommendations

- Optimize the cart page and simplify the checkout process to reduce cart abandonment.
- Increase investment in Email Marketing and Organic Search channels, which demonstrate the highest conversion rates.
- Implement automated email sequences triggered within 48 hours of a user's first visit.
- Continuously monitor AOV by traffic source to refine pricing and promotional strategies.
- Analyze seasonal revenue peaks to proactively plan inventory levels and server capacity requirements.